

CUSTOMER INSIGHT RECOMMENDATION SYSTEM

- **Data Driven Sales Analytics Dashboard**
- **Customer Behaviour Analysis**
- **Automated Business Insights Generation**
- **neeche:**
Python | Power BI | Machine Learning)

PROJECT OVERVIEW

Objective

- Analyze customer purchasing behaviour
- Identify sales patterns & trends
- Generate business recommendations

Solution

- Interactive dashboard
- Automated EDA analysis
- Data-driven decision support system

PROBLEM STATEMENT

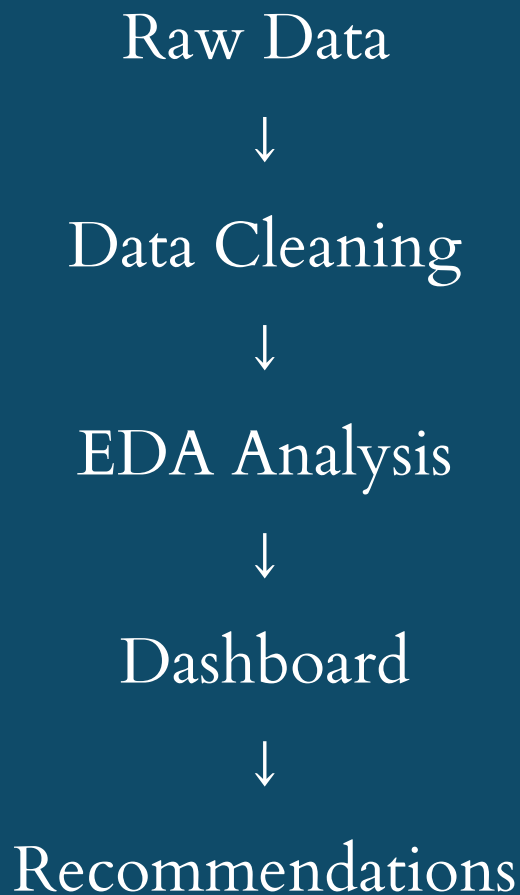
- Businesses generate large amounts of sales data
 - Manual analysis is time-consuming
 - Hidden customer trends remain unnoticed
 - Lack of insights affects decision making

Our system converts raw data into actionable insights

DATASET PROCESSING

- Online Retail II Dataset
- Transaction based customer records
- Invoice, Product, Quantity & Revenue details.

Workflow:



TECHNOLOGIES USED

TECH STACK

Python

- Pandas
- Matplotlib

Visualization

- Power BI
- Interactive Charts

Tools

- GitHub
- VS Code

DATA CLEANING & EDA

DATA PREPROCESSING

- Removed missing customer IDs
- Removed cancelled invoices
- Handled invalid quantities
- Created revenue features

EDA Analysis

- Sales Trends
- Customer Behaviour
- Product Performance

DASHBOARD FEATURES

INTERACTIVE DASHBOARD

- Revenue Overview
- Top Selling Products
- Customer Segmentation
- Sales Performance Tracking
- Category-wise Analysis



KEY INSIGHTS

BUSINESS INSIGHTS GENERATED



High Performing Areas

- Identified top revenue products
- Found best customer segments



IMPROVEMENT AREAS

- Low performing products detected
- Inventory optimization suggestions

RECOMMENDATION SYSTEM

SMART RECOMMENDATIONS

System suggests:

- Increase stock of high-demand products
- Focus on profitable categories
- Improve low sales areas
- Target valuable customers

CONCLUSION

RESULT

- ✓ Automated analysis system
- ✓ Faster business decisions
- ✓ Interactive visualization
- ✓ Data-driven growth strategy

Future Scope

- AI based forecasting
- Real-time analytics integration